

What makes a chord progression liked — findings

Generated by the analysis script `discover.py`, with a fixed random seed so the numbers reproduce exactly. Re-run the script to regenerate.

Summary of what was found

- **11 of 15** candidate patterns survived confirmation on the held-out half, kept their significance after the false-discovery-rate correction, **and** held a stable effect across bootstrap resamples.
- **The dominant driver is dissonance, or roughness**: progressions containing **fewer clashing note-pairs** — notes a semitone or less apart, sounding together — are reliably **liked**, while clash-heavy ones are **disliked**. This is the strongest signal found, and it comes from pure subtraction between note values, with no music theory imposed.
- Secondary, weaker signals: **smaller jumps between chords and a calmer top note**, and a **higher-pitched final chord**.
- **There is real signal**. A classifier separating liked from not-liked scores **0.68** where 0.50 is pure chance, against a shuffled-rating baseline of 0.50 ($p = 0.032$). Modest, but above chance.
- Effect sizes per pattern remain **small** (about 0.12 to 0.19 on the -1 to $+1$ scale): useful as *tendencies*, not as a confident per-progression predictor.

Strongest confirmed pattern

at most 2 near-unison clashes (two notes less than a semitone apart) → toward LIKE
Effect size +0.18 (95% interval [+0.07, +0.29]), p-value 0.0013, matching 186 progressions.

All confirmed patterns (significant after correction, with a stable effect)

pattern	direction	number matching	effect size	95% interval	p- value
at most 2 near-unison clashes (two notes less than a semitone apart)	LIKE	186	+0.18	[+0.07, +0.29]	0.0013
more than 3 semitone clashes (4 or more) AND more than 2 near-unison clashes (two notes less than a semitone apart) (3 or more)	DISLIKE	90	-0.19	[-0.30, -0.07]	0.0030
at most 3 semitone clashes AND at most 2 near-unison clashes (two notes less than a semitone apart)	LIKE	169	+0.16	[+0.04, +0.26]	0.0040
at most 3 semitone clashes	LIKE	213	+0.15	[+0.04, +0.27]	0.0143
more than 3 semitone clashes (4 or more)	DISLIKE	107	-0.15	[-0.26, -0.05]	0.0147
at most 2 near-unison clashes (two notes less than a semitone apart) AND the total movement of the top note across the chords is low (≤ 9)	LIKE	93	+0.15	[+0.03, +0.27]	0.0157
at most 2 near-unison clashes (two notes less than a semitone apart) AND the total distance the top voice travels is low (≤ 9)	LIKE	93	+0.15	[+0.01, +0.28]	0.0177
at most 3 semitone clashes AND the total distance the top voice travels is low (≤ 9)	LIKE	108	+0.15	[+0.03, +0.26]	0.0217
the final chord's 3rd note (pitch height) is high (> 64)	LIKE	141	+0.12	[+0.01, +0.23]	0.0233
the final chord's top note (pitch height) is high (> 67)	LIKE	134	+0.13	[+0.01, +0.24]	0.0240
at most 3 semitone clashes AND the total movement of the top note across the chords is low (≤ 9)	LIKE	108	+0.15	[+0.03, +0.26]	0.0243

Candidates that did NOT survive

Found in discovery but failed held-out confirmation — treat as noise:

- the biggest single jump any voice makes between chords is low (≤ 7) (p-value 0.068)
- at most 5 close clashes (two notes within a whole tone) (p-value 0.255)
- the total movement of the top note across the chords is low (≤ 11) (p-value 0.306)
- the total distance the top voice travels is low (≤ 11) (p-value 0.306)

All 15 tests, grouped by signal (confirmed + failed)

Every candidate carried from the discovery half into held-out permutation testing. Within each group, ordered strongest-first by p-value.  = survived the false-discovery-rate correction with a stable effect;  = failed.

Group A — Dissonance / roughness

pattern	effect size	p-value	verdict
at most 2 near-unison clashes (two notes less than a semitone apart)	+0.18	0.0013	
more than 3 semitone clashes (4 or more) AND more than 2 near-unison clashes (two notes less than a semitone apart) (3 or more)	-0.19	0.0030	
at most 3 semitone clashes AND at most 2 near-unison clashes (two notes less than a semitone apart)	+0.16	0.0040	
at most 3 semitone clashes	+0.15	0.0143	
more than 3 semitone clashes (4 or more)	-0.15	0.0147	
at most 2 near-unison clashes (two notes less than a semitone apart) AND the total movement of the top note across the chords is low (≤ 9)	+0.15	0.0157	
at most 2 near-unison clashes (two notes less than a semitone apart) AND the total distance the top voice travels is low (≤ 9)	+0.15	0.0177	

pattern	effect size	p-value	verdict
at most 3 semitone clashes AND the total distance the top voice travels is low (≤ 9)	+0.15	0.0217	✓
at most 3 semitone clashes AND the total movement of the top note across the chords is low (≤ 9)	+0.15	0.0243	✓
at most 5 close clashes (two notes within a whole tone)	+0.07	0.2552	✗

Group B — Voice-leading / motion

pattern	effect size	p-value	verdict
the biggest single jump any voice makes between chords is low (≤ 7)	+0.13	0.0676	✗
the total movement of the top note across the chords is low (≤ 11)	+0.07	0.3062	✗
the total distance the top voice travels is low (≤ 11)	+0.07	0.3062	✗

Group C — Final-chord height / register

pattern	effect size	p-value	verdict
the final chord's 3rd note (pitch height) is high (> 64)	+0.12	0.0233	✓
the final chord's top note (pitch height) is high (> 67)	+0.13	0.0240	✓

Combined conditions mostly add up rather than multiply. A formal test for interaction (comparing a model with a combined term against one without) found **no significant interaction** for the pairs tested — a two-part pattern raises the average rating by stacking two independent effects (less dissonance plus calmer movement between chords), not through genuine synergy. The practical takeaway is simply: **several weak signals accumulate.**

What the musical quantities mean

- **Semitone clashes** — how many pairs of notes sounding together sit one semitone apart. Fewer clashes means less dissonance, which is the strongest single predictor of being liked.
- **Near-unison clashes** — pairs of notes less than a semitone apart, close enough to beat against each other. Fewer is liked.
- **Tightest interval** — the smallest gap between any two notes anywhere in the progression; a very small gap sounds harsh.
- **Biggest single jump** — the largest distance any one voice moves when the chord changes. Smaller jumps make a smoother progression.
- **Top-note movement** — how far the highest note travels from chord to chord; less movement reads as calmer.
- **Final-chord height** — the pitch of the notes in the last chord; higher means a higher-pitched ending.

How consistent were the ratings? (the ceiling on any result)

measure	value
average disagreement with the 5 most similar progressions	0.544
how often near-identical progressions got different ratings	0.475
accuracy from always guessing the most common rating	0.610
estimated best accuracy any method could reach	0.456